2026 Sagan Summer Workshop

Exoplanets with Roman Surveys: Microlensing and Transits

Getting Started Guide: Google Colab

For the workshop hands-on sessions, the notebooks have been designed to run in Google Colaboratory. Users who are proficient in Python can download them and adapt them to run them as Python Jupyter notebooks, but only Google Colaboratory is fully supported. **In addition to following these instructions, please also run the Setup notebook (Hands-On Session I) and make a shortcut to the shared drive (Hands-On Session II) as described below.**

Google Colaboratory allows you to execute Python in a browser without configuring Python on your local system. The Python code is run from a notebook environment similar to Jupyter notebooks, with execution and text cells. For a general introduction to Colaboratory, see:

What is Colaboratory?
https://colab.research.google.com/notebooks/intro.ipynb

Overview of Colaboratory Features
https://colab.research.google.com/notebooks/basic_features_overview.ipynb

If you are new to Python, see Introduction to Python Programming document by Tim Brandt(STScI).

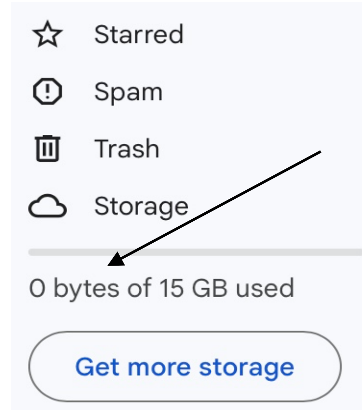
Workshop Colaboratory Instructions

You will need:

- A free Google account which includes 15GB storage. https://www.google.com/account/about/
- Having 5GB of free space should be sufficient. If you do not have 5GB available or wish to keep your personal Google account separate from the workshop activities, we suggest creating a new Google account to maximize storage.

Verify that you have about 5 GB storage available:
1) Log into your Google account
2) Navigate to Google Drive by either following this link: https://drive.google.com/drive/my-drive or, from your account, click on the dot navigation and then click on the Drive icon.

3) Once in Drive, on the left side menu, there is a section called "Storage" which will show how much space you have available. If you have insufficient storage in your Google account, we suggest creating a new account rather than purchasing storage. Note that if you reach the storage limit, it will affect your email usage, so it is better to create a new account.



SSW 2026 Notebooks

Copy the Notebooks individually to your Google Drive (see the **Steps to copy the Notebooks to your Google Drive** section below) or download the notebooks all at once and copy them into the "Colab Notebooks" directory in your Google Drive (or onto your local machine for Python Jupyter notebook usage) via the tar or zip format file.

Tar format: https://cosmos.ipac.caltech.edu/ssw/hands-on/SSW2026_notebooks.tar.gz
Zip format: https://cosmos.ipac.caltech.edu/ssw/hands-on/SSW2026_notebooks.zip

Hands-on Session I: Microlensing Fits of Single and Binary Lenses

Run the Setup Notebook
SSW2026_HandsOnI_Microlens_Setup.ipynb
https://colab.research.google.com/drive/1sUvWImFHpWnOW8zY-YivW7wXvDr5E8AK?usp=sharing
This needs to be run only **once** *and applies to all the Session I notebooks below.*

SSW2026_HandsOnI_SingleLens.ipynb
https://colab.research.google.com/drive/1zIexIlmOhLsVoitf27f7qMJSMoUzxEVr?usp=sharing
https://cosmos.ipac.caltech.edu/ssw/hands-on/SSW2026_Hands-OnI_Single_Lens_Appendix.pdf
Single Lens notebook and appendix

SSW2026_HandsOnI_Binary_Lens.ipynb
https://colab.research.google.com/drive/1Gk_rlDhcsbOk9l2jM8XFMDAdtanP4re4?usp=sharing
Binary Lens notebook

SSW2026_Hands-OnI_Group_Projects.pdf
https://cosmos.ipac.caltech.edu/ssw/hands-on/SSW2026_Hands-OnI_Group_Projects.pdf
Group Projects Description. Note for Group Project 2, follow the process of loading the data as described, but the list of files are available here:
https://github.com/rges-pit/sagan_session1/tree/main/Binary-Lens-Fitting/Binary-Events/data

SSW2026_HandsOnI_Group_Project.ipynb
https://colab.research.google.com/drive/1bMItb9SLaH1oN_TI_eYtz9oRv6HX9Oka?usp=sharing
Group Project 1

Hands-on Session II Placing Transit or Microlensing Events in a Galactic Context

Shared Drive: 2026 Sagan Activity - Galactic Context
You need to make a shortcut to this directory from the top level of your Google Drive:

1. Login to the Google account you plan to use for this activity.
2. Click the link to open the folder in your browser:
   https://drive.google.com/drive/folders/1gJhl7zA3lqkiXx3YnAdgU8Q14tUexLlm?usp=sharing_eil&ts=69fcd6a4$0
3. To make the shortcut **EITHER:**

   3a. In the top-right corner of the browser, check if there is an **"Add shortcut to Drive" icon** (it looks like a Google Drive folder with a plus sign). Click it and select **My Drive** as the destination.

   OR

   3b. Click on the **down triangle** next to the folder name, “Shared with me > 2026 Sagan Activity – Galactic Content”. Select **Organize,** then **Add shortcut**. Click the **All Locations** and then select **My Drive** as the destination and then “**Add**” .

SSW2026_HandsOnII_Overview_and_Group_Projects.pdf
https://cosmos.ipac.caltech.edu/ssw/hands-on/SSW2026_Hands-OnII_Overview_and_Group_Projects.pdf
Overview and Group Projects Description.

SSW2026_HandsOnII_ML_or_Transit_Context.ipynb
https://colab.research.google.com/drive/1x8AqPDS2efonGz8zGh72MKSwDRZ7I5es?usp=sharing
Transit or Microlensing Event in Context main exercise notebook

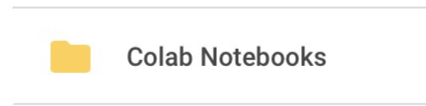
SSW2026_HandsOnII_Group_Project_GC_Event.ipynb
https://colab.research.google.com/drive/1M0paSJMfFJkRq4hp5vIANqJ6xMWrfW8V?usp=sharing
Group Project 1: Galactic Center (GC) Event

SSW2026_HandsOnII_Group_Project_Dark_Lens.ipynb
https://colab.research.google.com/drive/1nwJ1_GnAJ4xCzJDo2E05l_GSSR7sxeGN?usp=sharing
Group Project 2: Dark Lens

SSW2026_HandsOnII_Group_Project_SED_Fitting.ipynb
https://colab.research.google.com/drive/1MDpCbpc5psXc9-Y6lXc0xCMHoIB1lsZM?usp=sharing
Group Project 3: SED Fitting – Recommended for Python Users, not Colab

SSW2026_HandsOnII_Group_Project_Binary_Event.ipynb
https://colab.research.google.com/drive/1VNjARjNWxgC_Ak3544RWONQLOSorHa7R?usp=sharing
Group Project 4: Binary Event

Steps to copy the Notebooks to your Google Drive:

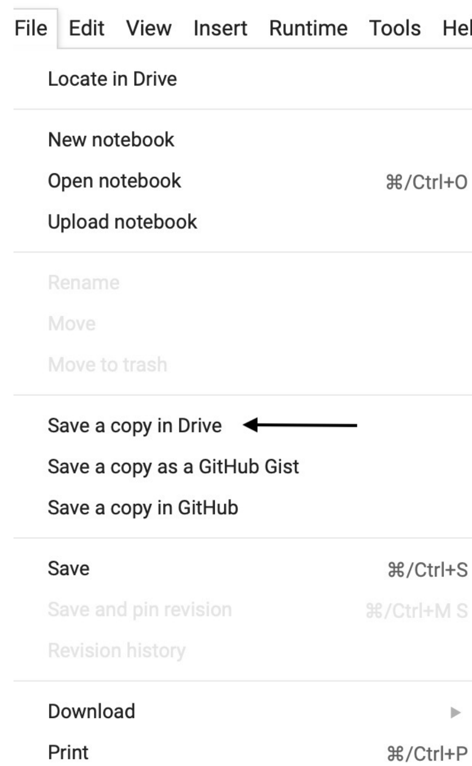


1) Log into your Google account with the available storage – make sure you have signed out of all other Google accounts if you have logged into more than one.

2) From the browser where you are logged into your Google account, enter the URL of the notebook (as shown above).

3) The notebook will open in your browser. You should see your profile/initial on the upper right side of the page. If you see "Sign in", log into your Google account.

4) Under the name of the notebook, you will see options like "File, Edit, View" etc. Click on "File" and then select "Save a Copy in Drive" (see image on the right).



5) You will be prompted to open the notebook in a new tab or window, and the notebook name will be prefaced with "Copy of". You can rename the notebook by clicking on its name. The notebooks will be saved in your Google Drive at https://drive.google.com/drive/my-drive in a directory called "Colab Notebooks".

6) Close the notebook browser windows.

7) Repeat this process to save all the Google Colab Notebooks to your Google Drive.

Instructions for Using Colab Notebooks

Launching the Colab Notebook

1) Log into your Google account where you have saved the notebooks.
2) Go to your Google Drive: https://drive.google.com/drive/my-drive
3) Click on Colab Notebooks directory.
4) Click on the notebook you want to work on.

Running the Colab Notebook

For the hands-on activity notebooks, *you should step through each cell individually* by clicking on the right-facing triangle to the left of each cell (▶). Be sure to run all the initialization cells before the exercise cells. Note that some cells may be marked **Colab only** or **Python only**; run only the **Colab** ones.

Useful Colab Top Menu items

- **File** -> Save Saves the file to your Google Drive
- **File** -> Download Downloads a .ipynb (Jupyter Notebook) to your local machine
- **Edit** -> Clear all outputs Clears the output from all cells
- **Runtime** -> Run all Run all the cells. Can be run multiple times.
- **Runtime** -> Disconnect and delete runtime Disconnects and exits the runtime, resetting the notebook variables back to its original state. This action is only useful if the notebook encounters an issue. It does not affect files downloaded to your drive. To restart the notebook, use the "Reconnect" button on the right.
- **Table of contents** – accessed via the three orange horizontal lines at the top left; clicking on them shows or hides the Table of contents.
- Closing the browser window stops the Colab instance.